

Equilibrium Transition and Cartel Formation

A Structural Analysis of Chile's Pharmacy Cartel

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How do firms start colluding?

- ▶ Firms must establish a common understanding of how to raise prices together.
- ▶ They must also decide whether rivals will participate in a proposed increase.
- ▶ Chile's pharmacy cartel lets me study these two problems during the transition from a price war to coordinated increases.

Knowing the procedure does not guarantee participation.

Two stages of coordinated price increases

- ▶ **Margin restoration:** raise former loss-leader prices enough to restore positive retail margins.
- ▶ **Rent extraction:** coordinate a further increase above those restored-margin prices.
- ▶ In the observed price paths, margin restoration spreads faster than rent extraction.

Why does the same procedure spread more slowly to the second increase?

An empirical model of cartel formation

- ▶ **Demand:** estimate how low prices attract customers before and after the advertisement ban.
- ▶ **Verification:** model how followers come to understand the price-increase procedure.
- ▶ **Participation:** let leaders update their assessment of subsequent increases while followers retain their own choices.

Related work: coordination and compliance (Harrington, 2018), learning through price leadership (Byrne and de Roos, 2019), and incentives to sustain collusion (Igami and Sugaya, 2022).

Three chains and a loss-leader incentive

- ▶ Cruz Verde, Farmacias Ahumada (FASA), and Salcobrand accounted for about 92% of national pharmacy sales in 2007.
- ▶ A low medicine price can attract a shopper who also buys other medicines or complementary products.
- ▶ Profit on the rest of that basket can make a below-cost medicine price worthwhile.

The relevant incentive includes both medicine margins and complementary basket profit.

Daily prices, quantities, and documentary evidence

- ▶ A processed panel of **222 medicines**, three chains, and daily observations in 2006–2008.
- ▶ Transaction prices and quantities measure realized price paths and demand. Listed-price schedules are a separate series.
- ▶ Case records establish the advertisement campaign, the ban, and laboratory-mediated communication.

Three price tiers organize the evidence

Tier 0 Below-cost prices during the price war.

Tier 1 The first common price tier that restores a positive retail margin: **margin restoration**.

Tier 2 The next common tier above Tier 1: **rent extraction**.

An **event** is one medicine-level increase attempt and the rivals' responses. Each medicine enters each first-crossing count at most once.

A direct $0 \rightarrow 2$ event crosses both boundaries but is one event.

From the price war to coordinated increases

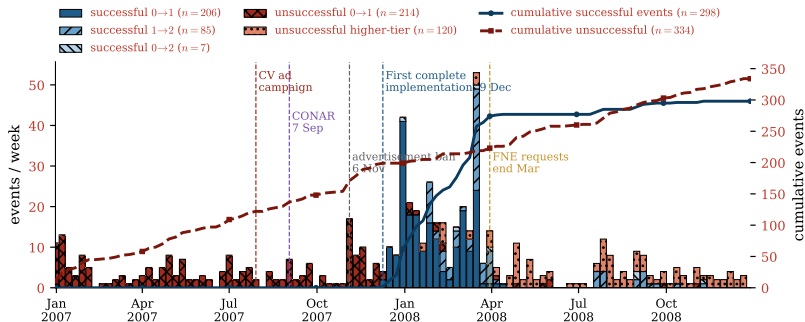
- August 2007 Cruz Verde runs a comparative advertisement campaign; further cuts follow.
- 6 Nov 2007 The binding advertisement ban ends the campaign; the sustained price war ends at the same time.
- 9 Dec 2007 First complete three-chain increase sequence.
- 27 Jan 2008 First Tier 1→2 increase.
- Late March 2008 FNE information requests precede the end of the cartel wave.

A common procedure takes time to establish

- ▶ Laboratories help arrange a repeatable sequence: one chain raises first, then the others decide whether to follow.
- ▶ From 6 Nov to 8 Dec 2007, there are **42 unsuccessful medicine-level attempts**; Salcobrand initiates 39.
- ▶ From 9 Dec 2007 to 31 Mar 2008, there are **273 successful events**; Salcobrand initiates 208.

First-mover identity records the observed order of price changes. The documentary record supplies evidence about knowledge of the procedure.

Failed attempts precede the coordinated wave



Full 2007–2008 panel: totals cover a wider window than the preceding slide.

Margin restoration spreads faster

First-crossing rate	Tier 0→1	Tier 1→2
Per 100 eligible medicine-weeks	9.83	5.39

The margin-restoration rate is **1.83 times** the rent-extraction rate in matched 65-day windows.

- ▶ Exposure changes daily as medicines enter and leave the risk set.
- ▶ The model must explain both the fast first wave and the slower second wave.

Windows: 9 Dec–11 Feb and 27 Jan–31 Mar; direct 0→2 events treated separately.

Demand: three chains and an outside good

Chain i , medicine j , day t : quantity q_{ijt} gives the shares

$$s_{ijt} = q_{ijt} / N_j, \quad s_{0jt} = 1 - \sum_i s_{ijt}.$$

- ▶ The outside good is not purchasing from any of the three chains.
- ▶ N_j : maximum total daily quantity on complete-observation days, divided by 0.92, plus one unit.
- ▶ Retain medicine-days with positive prices and quantities at all three chains.

Market size is a maintained normalization.

Demand responds to position and recent price changes

- ▶ **Low-price group** L_{ijt} : price is at least 5% below the highest chain price; up to two chains qualify.
- ▶ **Recent increases** A_{ijt}^+ **and cuts** A_{ijt}^- : cumulative positive and negative log changes over the current and preceding six days.
- ▶ **Unmatched high price** H_{ijt} : a recent increase leaves both rivals at least 5% below the chain's price. Its log gap disappears when a rival matches.

The history variables measure magnitudes, not event indicators.

A daily logit demand equation

$$\log(s_{ijt}/s_{0jt}) = \phi_{ij} + \zeta_t - \alpha_d p_{ijt} + \lambda_d L_{ijt} \\ + \gamma_{\text{inc}} A_{ijt}^+ + \gamma_{\text{cut}} A_{ijt}^- + \gamma_H H_{ijt} + \xi_{ijt}.$$

- ▷ d selects pre- or post-ban coefficients; price p is in thousand CLP.
- ▷ ϕ_{ij} and ζ_t : chain–medicine and common-day fixed effects; ξ_{ijt} : residual demand shock.
- ▷ The price and low-price-group coefficients may change after the ban; history coefficients are common.

Identification: listed prices instrument paid prices

- ▶ National listed prices are chain schedules against which transaction discounts are set.
- ▶ **Maintained exclusion:** conditional on fixed effects and included controls, listed prices affect demand only through transaction prices.
- ▶ Their timing motivates this assumption: the national schedule is set before the residual transaction-stage demand shock.

589,624 chain–medicine–day observations; standard errors clustered by medicine.

First-stage partial R^2 : .953 pre-ban, .965 post-ban.

The estimated low-price reward falls after the ban

	Pre-ban	Post-ban
Price coefficient α	.034 (.006)	.028 (.005)
Low-price-group coefficient λ	.207 (.016)	.062 (.014)
Mean own-price elasticity	−.353	−.314

Low-price membership raises purchase odds relative to the outside good by **23.0%** before the ban and **6.4%** afterward.

Parentheses: medicine-clustered SEs; price in thousand CLP.

Medicine margins and complementary basket profit

Let $\tilde{q}_{ij,t}$ be fitted daily quantity at the chosen prices.

$$\bar{\pi}_{ij,t} = \frac{\tilde{q}_{ij,t}}{\tilde{N}} (p_{ij,t} - c_j + \mu).$$

- ▶ $\tilde{N}$ is median medicine market size; c_j is median wholesale cost, common across chains and time invariant.
- ▶ μ is complementary basket profit per customer-equivalent unit, measured in thousand CLP.
- ▶ The same μ enters all blocks within a specification. The ban changes fitted demand at low prices.

daily quantity normalized to compare medicines.

From daily profit to the weekly choice payoff

$$\pi_{ij,t} = 7\bar{\pi}_{ij,t} + 7\beta\Delta\bar{\pi}_{ij,t+1}.$$

- ▶ Current action prices apply for seven days.
- ▶ $\Delta\bar{\pi}_{ij,t+1}$ is the next week's profit difference due to its opening price change, evaluated at the same next-week prices with and without that history.
- ▶ This history correction lasts one week. Future value enters separately through the Bellman continuation.

Weekly discount factor $\beta = 0.80^{1/52}$.

Two games, with a campaign block inside Game 1

Game 1: before campaign Weekly review probability ω_1 ; ordinary cut depths.

Game 1: campaign Weekly review probability ω_2 ; medicine-specific cut depths calibrated to observed prices.

Game 2: after ban Post-ban demand and reviews tied to observed laboratory dates and medicine counts.

The model assigns the Game 2 switch to 4–10 Nov 2007, the week containing the ban. Cut and increase choices remain endogenous.

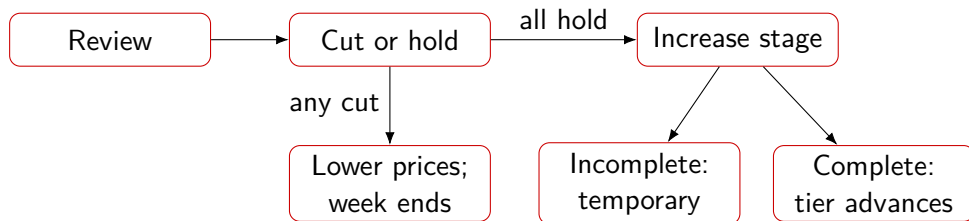
Game 1: 31 Dec 2006–3 Nov 2007; campaign begins 5 Aug in the model.

Persistent prices and the economic state

- ▶ Medicine j has price state z_{jt} : initial prices l , price-war levels $0^{(0)}, \dots, 0^{(K)}$, or Tier 1 or 2.
- ▶ A complete three-chain increase advances the persistent tier. An incomplete increase lasts the current week only.
- ▶ Verification R_t records common knowledge of the procedure. Once one, it remains one.
- ▶ A cut made with $R_t = 1$ starts a separate punishment continuation next week, with no further increase opportunities for that medicine.

Time t now indexes model weeks.

At a review: cut first, then consider an increase



Game 1: simultaneous increase choices. Game 2 after verification: a leader followed by sequential participation decisions.

This sequence describes reviews outside punishment; additional pre-verification cuts can occur outside reviews.

Before verification: different understandings of play

- ▶ Salcobrand knows the laboratory procedure and is the only candidate leader for Tier $0 \rightarrow 1$.
- ▶ Uninformed followers solve the decentralized campaign-block game under post-ban demand. They expect others to play that game and do not anticipate the procedure.
- ▶ Each unverified week ends with probability η_0 of common verification; informed play starts the following week.

Verification is a modeled information transition, dated against the first complete sequence.

After verification: followers still decide whether to join

- ▷ Each of the six orders of the three chains is equally likely. In that order, the first chain accepting the lead becomes leader.
- ▷ If everyone waits, there is no increase. Otherwise, the other two chains decide sequentially whether to follow.
- ▷ Knowing the procedure makes these decisions well defined; it does not make complete participation certain.

Both tiers retain follower decisions in both specifications.

What differs between the two confidence models?

For a verified rent-extraction proposal, m_t^r is the leader's subjective weight on the value of sequential follower choices.

Unit Confidence Fix $m_t^r = 1$: evaluate leading using the sequential follower outcome law.

Adaptive Confidence Update m_t^r after complete and incomplete Tier 1→2 attempts.

Actual followers make their own decisions in both models. The weight is not the probability that both follow.

The leader weighs sequential participation against a solo raise

For a verified Tier 1→2 proposal,

$$\bar{v}_{ij,t}^{\text{lead}} = v_{ij,t}^{\text{solo}} + m_t^r (v_{ij,t}^{\text{seq}} - v_{ij,t}^{\text{solo}}).$$

- ▷ v^{solo} : weekly profit and continuation if both rivals hold.
- ▷ v^{seq} : weekly profit and continuation averaged over the followers' sequential choices.
- ▷ Verified Tier 0→1 uses v^{seq} directly; only rent extraction uses the subjective weight.

Incomplete participation can leave the leader uniquely high priced.

Payoff differences determine action probabilities

At a reached decision node h , action 1 is cut, lead, or follow; action 0 is hold, wait, or decline.

$$\Pr(a_i = 1 \mid x_{jt}, h) = \Lambda\left(\frac{v_{ij,t}^h(1; x_{jt}) - v_{ij,t}^h(0; x_{jt})}{\tau}\right).$$

- ▶ v^h includes current payoff and discounted continuation; x_{jt} is the economic state.
- ▶ $\Lambda(u) = 1/(1 + e^{-u})$; the action scale τ is fixed.
- ▶ A follower anticipates later choices. Random shocks do not force it to participate.

Adaptive updating uses rent-extraction outcomes

Let S_t^r and F_t^r count complete and incomplete initiated Tier 1→2 attempts before week t .

$$m_t^r = \frac{\nu^r m_0^r + S_t^r}{\nu^r + S_t^r + F_t^r}.$$

- ▷ m_0^r : initial weight; ν^r : effective prior count.
- ▷ A complete attempt raises the weight; an incomplete attempt lowers it.
- ▷ This is a Beta-shaped outcome rule for a payoff weight. Tier 0→1 outcomes do not update it.

Updating begins only after verification.

One medicine's outcome affects later choices for others

The public information state and the medicine's economic state are

$$G_t = (R_t, S_t^r, F_t^r), \quad x_{jt} = (z_{jt}, G_t).$$

- ▶ Every medicine selected in week t uses the same opening counts.
- ▶ Complete and incomplete outcomes are aggregated when the week closes; they affect next week's weights.
- ▶ Punishment remains separate. Learning on other medicines does not reopen a punished medicine's increase opportunities.

Estimation: fit prices, timing, and both increase waves

- ▶ Estimate each specification separately using 100 simulated paths per candidate.
- ▶ Nine loss blocks: two tier-count paths, three price paths, terminal war depth, starting Game 2 price, state shares, and verification timing.
- ▶ Within each block, average squared standardized model–data discrepancies; then average the nine block losses equally.
- ▶ Exclude Game 1 cut counts: observed screened events and all simulated cut actions have different coverage.

Fixed calibrated campaign cut depths and observed laboratory schedule.

Conditional economic parameter estimates

	Unit	Adaptive
Basket profit μ	16.328 (.590)	15.820 (.539)
Pre-campaign review ω_1	.02328 (.00113)	.02344 (.00105)
Campaign review ω_2	.10500 (.00797)	.10375 (.00837)
Verification η_0	.11776 (.02076)	.19839 (.02045)

μ : thousand CLP per customer-equivalent unit; other rows: weekly probabilities (η_0 conditional on no verification).
 Parentheses: bootstrap SEs in the same units.

Adaptive estimates an initial weight and prior strength

	Unit	Adaptive
Initial weight m_0^r	Fixed at 1	.000725 (.02011)
Prior strength ν^r	–	375 (90.56)

The standardized nine-block loss is **6.852** for Unit and **3.131** for Adaptive. Each specification is estimated separately. Their fitted-path difference does not isolate the subjective-weight rule alone.

Prior strength is an effective count; parentheses are bootstrap SEs.

Adaptive is closer at the final week

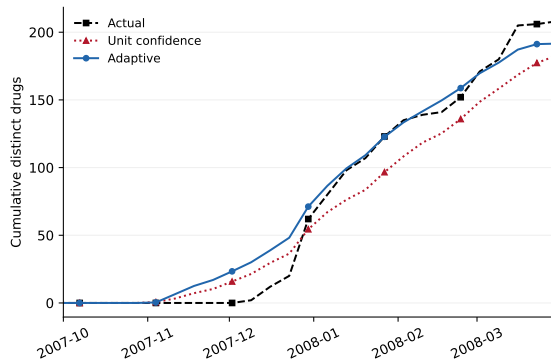
Cumulative first-crossing medicine counts, 30 Mar–5 Apr 2008:

	Data	Unit	Adaptive
Margin restoration (0→1)	208	182.61	191.58
Rent extraction (1→2)	70	106.22	67.67

- ▶ Unit produces too many second increases by the endpoint.
- ▶ Both specifications begin the first wave too early, as the next figure shows.

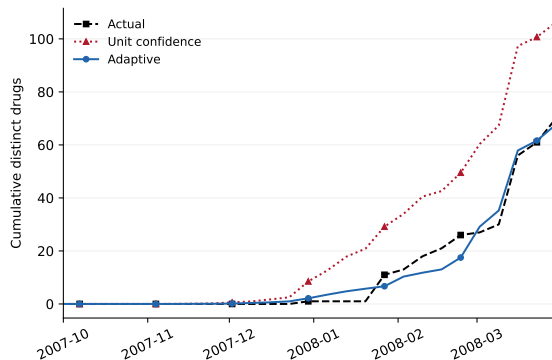
Simulation counts average 100 paths; each medicine enters each count at most once.

Margin restoration begins too early in both models



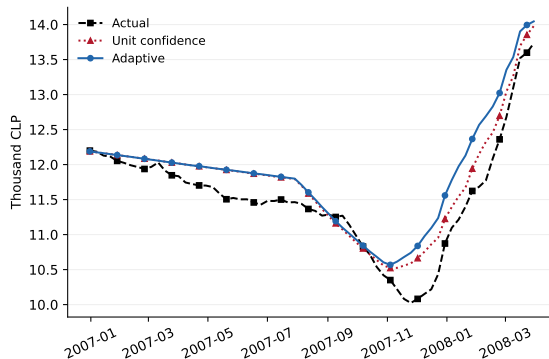
Cumulative medicines crossing 0→1; model lines average 100 paths at separately estimated parameters.

Adaptive more closely matches the rent-extraction path



Cumulative medicines crossing 1→2; model lines average 100 paths at separately estimated parameters.

Average prices recover, with remaining fit differences



Equal-weight medicine-chain weekly mean prices, thousand CLP; simulation lines average 100 paths.

What the estimates are conditional on

- ▷ Approximate inner-game policies, with a cap of 30 iterations.
- ▷ One damped pre-estimation update of Adaptive's learning transition, held fixed during the search; no converged transition fixed point is claimed.
- ▷ Missing Tier 2 price benchmarks filled using a positive-markup regression.
- ▷ 100 successful paired laboratory-block bootstrap replications for the reported standard errors.

Conclusion: knowing the procedure is only the first step

- ▶ The estimated low-price reward falls after the advertisement ban, weakening the loss-leader incentive.
- ▶ In the model, verification makes the procedure common knowledge while followers retain participation choices.
- ▶ Adaptive Confidence better matches the slower rent-extraction path at separately estimated parameters, subject to the stated computational scope.

Thank you

Questions and discussion

Backup: one-off demand and profit effects

- ▶ Median quantity loss from an unmatched initial-to-Tier 1 raise: **12.2% pre-ban, 9.8% post-ban.**
- ▶ Median quantity gain from the initial five-percent cut: **2.3% pre-ban, 1.5% post-ban.**
- ▶ For the raise, the median plotted profit-change curve crosses zero at $\mu = 3.40$ pre-ban and 4.90 post-ban (thousand CLP per customer-equivalent unit).

Fixed two-week paths, excluding continuation. Quantity percentages: first-week change plus next-week history correction, divided by two-week no-history baseline sales.

Backup: price-state transitions

$$z_{j,t+1} = \begin{cases} 0^{(0)}, & \text{first cut with } R_t = 1 : \text{ enter punishment,} \\ \mathcal{D}_{jt}(z_{jt}), & \text{other cut,} \\ \mathcal{U}(z_{jt}), & \text{no cut and a complete increase,} \\ z_{jt}, & \text{otherwise.} \end{cases}$$

- ▷ $\mathcal{D}_{jt}$ applies the ordinary or calibrated cut depth.
- ▷ $\mathcal{U}$ sends initial/price-war states to Tier 1 and Tier 1 to Tier 2.
- ▷ Later punishment cuts follow $\mathcal{D}_{jt}$; no increase opportunities return.

Branches describe eligible actions; information at the opening of the week determines punishment.

Backup: review opportunities are not guaranteed increases

- ▶ At laboratory dates, randomly select eligible medicines up to observed batch counts.
- ▶ Add supplementary verified Tier 0→1 opportunities so random allocation does not itself exclude margin restoration.
- ▶ No supplementary Tier 1→2 opportunities. Followers may decline any reviewed increase.
- ▶ Before verification, the simulation also conditions additional cuts on observed weekly totals.